

Deimonas Suchanka

Validation Quality Engineer | The Hague, Netherlands

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Analytical Systems Engineer who solves complex electro-mechanical problems through data-driven root cause analysis. My work has consistently involved engineering new processes and technical frameworks that deliver significant, quantifiable improvements in system performance and reliability.

KEY ENGINEERING PROJECTS

- **Pekos System RCA:** Diagnosed gain instability; improved force consistency by +350%.
- **QA Validation Framework:** Architected ISO 9001-aligned framework SOP for knowledge capture.
- **"Docs-as-Code" Workflow:** Implemented Git/LaTeX workflow as single source of truth for manuals.

PROFESSIONAL EXPERIENCE

Ventil Test Equipment BV

Service Engineer (Systems Process Focus)

Rijswijk, The Netherlands

November 2024 – Present

- Executed a data-driven RCA on a critical clamping system, improving production consistency by +350%.
- Architected and implemented a "docs-as-code" workflow (Git, LaTeX) to serve as a single source of truth.
- Designed a tiered QA validation framework to formalize knowledge capture in line with ISO 9001 principles.

Fast Fluid Management

Assembly Technician

Sassenheim, Netherlands

May 2024 – October 2024

- Assembled, tested, and troubleshoot complex mechanical and electrical dispensing systems independently.
- Performed quality control checks to ensure all machines met operational standards before shipping.

CERTIFICATIONS & CLEARANCES

VOL-VCA (Safety for Operational Supervisors)

EU/NATO Citizen, Lithuanian Armed Forces Veteran (2020-2021), Eligible for Security Clearance

CORE TOOLS

Git • MkDocs • LaTeX • progeCAD • Data-Driven RCA • ISO 9001 • VFD/PLC Tuning

EDUCATION

University of Amsterdam

Coursework toward Bachelor of Arts, European Studies

Amsterdam, Netherlands

2021 – 2022

- Developed strong academic writing and research skills.

Root Cause Analysis: Pekos Clamping System

Appendix: Case Study A

+350% Improvement in Force Consistency

Diagnosed a critical system instability and deployed a control solution that became the blueprint for data-driven analysis in the workshop.

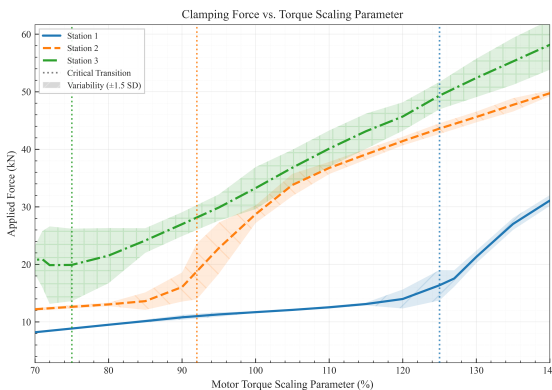
THE PROBLEM

A key clamping system exhibited severe force inconsistency, causing quality issues and line stoppages. My objective was to diagnose and resolve the instability using empirical data, as prior attempts had failed.

METHODOLOGY

- Data Acquisition:** Deployed sensors to capture real-time performance data and establish a quantitative baseline.
- Data Analysis:** Revealed a critical PID loop gain margin instability via time-series analysis in Python.
- Solution Development:** Engineered and validated a revised set of PID tuning parameters after modeling system dynamics.
- Deployment:** Implemented the validated parameters on the production machine and authored a new calibration SOP.

DATA VISUALIZATION



(Click image to view high-resolution version)

DELIVERABLES

Engineering Report	An 11-page IEEE-formatted report detailing the full RCA, methodology, and validated results.
QA Framework	A strategic proposal and lean SOP for formalizing knowledge capture under ISO 9001.
VFD Parameter Set	A finalized master list of station-specific inverter parameters to ensure stable machine operation.

Architecture: Tiered QA Validation Framework

Appendix: Case Study B

From Ad-Hoc to Standardized QA

Architected a new validation framework to move from basic functional checks to objective, data-driven quality assurance, addressing senior management’s goal for improved knowledge capture.

THE PROBLEM

Existing quality checks verified basic functionality but consistently missed subtle performance variations. This created three core business problems: a lack of objective QA data, no formal process for capturing engineering knowledge, and a gap in ISO 9001 alignment.

METHODOLOGY

- Foundational Analysis:** Leveraged findings from the Pekos system analysis to demonstrate the need for a more rigorous, data-driven validation process.
- Workflow Modeling:** Analyzed the existing ad-hoc workflow and designed a new, structured process with clear decision points and escalation paths.
- Framework Architecture:** Designed a tiered (L1-L3) system to efficiently handle routine QA checks as well as deep-dive root cause analysis for complex issues.
- Metric & Procedure Definition:** Defined specific, data-driven performance metrics (e.g., Consistency Index) and developed a lean, actionable SOP for Level 1 validation.

FRAMEWORK ARCHITECTURE

- Level 1: Routine Validation** | A lean, standardized procedure for commissioning engineers to perform data-driven pass/fail checks on all machines.
- Level 2: Targeted Diagnosis** | A focused investigation, triggered by a Level 1 failure, to quickly diagnose the issue and determine a corrective action.
- Level 3: Deep-Dive RCA** | A full, systematic root cause analysis involving a cross-functional team for complex issues, modeled on the Pekos investigation.

DELIVERABLES

QA Framework Proposal	A formal proposal outlining the tiered (L1-L3) framework, stakeholder impact, key metrics, and implementation plan.
Level 1 Test Procedure	A lean, single-page SOP for conducting routine performance validation in the workshop.
Data Template	An Excel template for data collection and automatic calculation of the Consistency Index (CI) metric during Level 1 tests.

**Note: These documents contain proprietary information and are available to be discussed in detail during an interview.*

System: "Docs-as-Code" Technical Documentation Workflow

Appendix: Case Study C

Foundation for Automated Documentation

Independently architected a modern documentation system from the ground up, replacing a chaotic file-based process with a version-controlled, single-source-of-truth workflow.

THE PROBLEM

The existing process for creating technical manuals was non-existent. It consisted of scattered Word documents on a shared drive with no version control, inconsistent formatting, and outdated information. This led to customer complaints and significant internal inefficiency.

METHODOLOGY

- Research Strategic Pivot:** Initially researched a complex pipeline for legacy '.docx' workflows, then pivoted to a more robust and pragmatic "docs-as-code" approach from first principles.
- System Development:** Established a complete development environment using VSCode, created a comprehensive style guide, and built out reusable templates through iterative experimentation.
- Implementation Delivery:** Authored new, high-quality technical manuals using the system, including taking ownership of custom technical illustrations by editing P&ID schematics in progeCAD.

SOLUTION ARCHITECTURE

- Version Control as Foundation:** Used Git for rigorous version control, providing a complete history of changes and establishing a single source of truth for all documentation.
- Dual-Output Workflow:** Architected two parallel pipelines: a LaTeX-based workflow for superior print quality, and a Markdown-based workflow for responsive, searchable static websites (using MkDocs).
- Foundation for Automation:** Designed the entire system with a "Lego block" philosophy, creating a modular foundation to enable the future automatic generation of machine-specific manuals from a given configuration.

DELIVERABLES

Documentation Workflow	A complete, version-controlled system for generating both print (LaTeX) and web (Markdown/MkDocs) docs from a structured process.
Style Guide & Templates	A comprehensive style guide and set of boilerplate templates to ensure all future documentation is consistent and professional.
Software Manual Prototype	A functional static website and self-contained '.exe' (via Tauri 2.0) for the CRS10 software manual, approved by the lead developer.

**Note: The system architecture and templates are proprietary and are available to be discussed in detail during an interview.*